Covariant Biquaternionic Tetrads in UBT: Central Anticommutator Metric and the Local Rank Theorem

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Abstract

We formulate the local classical geometry of Unified Biquaternion Theory without an embedding map, a compact-fiber average, or an externally chosen projection. The covariant first jet $E_\mu := \mathcal{N}_0^{-1/2} D_\mu \Theta$ is restricted in the classical sector to a real Lorentz slice of the biquaternions. On this slice the symmetrised product with quaternion conjugation is automatically central and equals a Lorentzian scalar times the algebra unit. This defines the metric intrinsically. We prove that the tetrad-to-metric differential has rank ten at every nondegenerate tetrad, with a six-dimensional kernel equal to local Lorentz frame rotations. The old comparison "eight real components of Θ versus ten metric components" therefore is not a valid local kinematic obstruction: the relevant object is the first covariant jet. The remaining problem is dynamical and integrability-based: derive the connection, preserve the Lorentz slice, and solve $D_\mu \Theta = E_\mu$ on shell.

1 Algebraic setting

Let

$$\mathbb{B} = \mathbb{C} \otimes \mathbb{H}$$

with commuting complex unit i and quaternion units $\mathbf{e}_j \mathbf{e}_k = -\delta_{jk}\mathbf{1} + \epsilon_{jk\ell}\mathbf{e}_\ell$. Denote quaternion conjugation, which reverses the three quaternion units but does not conjugate the commuting complex coefficient, by $\sharp$:

$$(z^0\mathbf{1} + z^k\mathbf{e}_k)^\sharp = z^0\mathbf{1} - z^k\mathbf{e}_k.$$

Definition 1.1 (Lorentz slice)**.** *The classical Lorentz module is the real four-dimensional subspace*

$$W_L := \{\, ix^0\mathbf{1} + x^k\mathbf{e}_k \mid x^a \in \mathbb{R} \,\} \subset \mathbb{B}.$$

For $X = ix^0\mathbf{1} + x^k\mathbf{e}_k$ and $Y = iy^0\mathbf{1} + y^k\mathbf{e}_k$, define

$$\eta(X, Y) := -x^0 y^0 + \sum_{k=1}^{3} x^k y^k.$$

Theorem 1.2 (Central Jordan identity)**.** *For all $X, Y \in W_L$,*

$$\boxed{\frac{1}{2}\left(X^\sharp Y + Y^\sharp X\right) = \eta(X,Y)\mathbf{1}.} \tag{1}$$

Thus the symmetrised product is already a real central biquaternion; no trace, real-part map, or fiber projection is required.

Proof. Write $X = (ix^0, \mathbf{x})$ and $Y = (iy^0, \mathbf{y})$ in scalar–vector quaternion notation, for which $(a, \mathbf{u})(b, \mathbf{v}) = (ab - \mathbf{u}\cdot\mathbf{v}, a\mathbf{v} + b\mathbf{u} + \mathbf{u}\times\mathbf{v})$. Then $X^\sharp = (ix^0, -\mathbf{x})$. The scalar part of $X^\sharp Y$ is $-x^0y^0 + \mathbf{x}\cdot\mathbf{y}$. Its vector part is $ix^0\mathbf{y} - iy^0\mathbf{x} - \mathbf{x}\times\mathbf{y}$. The vector part of $Y^\sharp X$ is its negative, while the scalar part is the same. Their half-sum is therefore exactly (1). □

2 Covariant first jet and metric

The fundamental UBT field remains $\Theta(q,\tau) \in \mathbb{B}$. On physical spacetime it is a smooth field. Let D_μ be the covariant derivative in the representation carried by Θ and let $\mathcal{N}_0 > 0$ be a fixed global unit-setting constant.

Definition 2.1 (UBT tetrad sector)**.** *Define*

$$E_\mu := \mathcal{N}_0^{-1/2} D_\mu \Theta. \tag{2}$$

The classical tetrad sector consists of configurations for which $E_\mu \in W_L$ and the four E_μ are linearly independent over $\mathbb{R}$. The metric is the unique real coefficient in the central identity

$$\boxed{\frac{1}{2}\left(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu\right) = g_{\mu\nu}\mathbf{1}.} \tag{3}$$

Writing $E_\mu = ie_\mu{}^0\mathbf{1} + e_\mu{}^k\mathbf{e}_k$ gives immediately

$$g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}, \qquad \eta_{ab} = \mathrm{diag}(-1, 1, 1, 1). \tag{4}$$

No local normalization is permitted; such a denominator would alter the field equations and can freeze a metric component. $\mathcal{N}_0$ is constant and may be absorbed into units.

Definition 2.2 (Algebraic bivector companion)**.** *The antisymmetric companion of the metric is*

$$\Sigma_{\mu\nu} := \frac{1}{2}\left(E_\mu^\sharp E_\nu - E_\nu^\sharp E_\mu\right), \qquad \Sigma_{\nu\mu} = -\Sigma_{\mu\nu}. \tag{5}$$

It contains oriented-plane/spin information. It must not be confused with the curvature $[D_\mu, D_\nu]$.

3 Local rank theorem

Let $e = (e_\mu{}^a)$ be a nondegenerate tetrad and define $\Phi(e) = e\eta e^\mathsf{T}$.

Theorem 3.1 (Surjectivity of the tetrad-to-metric differential)**.** *At every invertible real tetrad, the differential*

$$d\Phi_e : \mathbb{R}^{4\times 4} \longrightarrow \mathrm{Sym}_4(\mathbb{R})$$

has rank ten. Its kernel has dimension six and consists precisely of infinitesimal local Lorentz rotations of the tetrad.

Proof. The first variation is

$$\delta g_{\mu\nu} = \eta_{ab}\left(\delta e_\mu{}^a e_\nu{}^b + e_\mu{}^a \delta e_\nu{}^b\right). \tag{6}$$

Given an arbitrary symmetric tensor $h_{\mu\nu}$, choose

$$\delta e_\mu{}^a = \frac{1}{2} h_{\mu\rho} e^{\rho a}, \tag{7}$$

where $e^{\rho a}$ is the inverse tetrad with its Lorentz index raised. Direct substitution into (6) gives $\delta g_{\mu\nu} = h_{\mu\nu}$. Hence $d\Phi_e$ is surjective and has rank ten. The domain has dimension sixteen, so the kernel has dimension six. Variations $\delta e = e\lambda$ with $\lambda^\mathsf{T}\eta + \eta\lambda = 0$ lie in the kernel and form the six-dimensional Lorentz Lie algebra; therefore they exhaust it. □

Corollary 3.2 (Resolution of the naive component-count obstruction)**.** *The local algebraic map $E_\mu \mapsto g_{\mu\nu}$ has full metric rank. The comparison $\dim_\mathbb{R} \mathbb{B} = 8 < 10$ is irrelevant to this map because the metric is determined by four covariant first derivatives, not by the value of Θ alone. This closes only the* kinematic tetrad rank *question.*

4 Connection, Christoffel symbols, and flat limit

A connection is the rule needed to compare internal frames at neighboring points. In representation-independent notation,

$$D_\mu\Theta = \partial_\mu\Theta + \rho_*(\Omega_\mu)\Theta, \tag{8}$$

where Ω_μ is a biquaternionic/Lorentz-frame connection and ρ_* specifies how it acts on Θ. This notation deliberately does not choose left, right, or adjoint action before the representation is fixed.

The Christoffel symbols $\Gamma^\rho{}_{\mu\nu}$ and Ω_μ are related but are not the same object. Γ transports coordinate indices; Ω transports the local biquaternionic/Lorentz frame. Their compatibility is the tetrad postulate

$$\boxed{\partial_\mu E_\nu - \Gamma^\rho{}_{\mu\nu} E_\rho + \rho_{\mathrm{vec}*}(\Omega_\mu) E_\nu = 0.} \tag{9}$$

For a chosen torsion condition and a nondegenerate tetrad, this equation makes Γ and Ω two descriptions of the same local geometry.

The connection curvature is

$$\mathcal{R}_{\mu\nu}(\Omega) := [D_\mu, D_\nu] = \partial_\mu\Omega_\nu - \partial_\nu\Omega_\mu + [\Omega_\mu, \Omega_\nu] \tag{10}$$

with the last expression understood in the selected representation. In a flat simply connected region, $\mathcal{R}_{\mu\nu} = 0$ permits a local frame in which $\Omega_\mu = 0$. In inertial Cartesian coordinates one also has $\Gamma^\rho{}_{\mu\nu} = 0$, and $D_\mu = \partial_\mu$: this is the special relativistic sector. In curvilinear coordinates Γ and Ω may be nonzero even when the curvature vanishes.

5 The remaining nontrivial bridge

The tetrad is not postulated independently; UBT requires $E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0}$. Therefore full closure needs more than the rank theorem. One must prove:

1. existence of a connection and field solving $D_\mu \Theta = \sqrt{\mathcal{N}_0} E_\mu$;
2. preservation of the Lorentz slice $E_\mu \in W_L$ and centrality of (3);
3. derivation or unique elimination of Ω_μ from the UBT action, so it is not an arbitrary extra field;
4. compatibility/integrability. In a torsion-free coordinate sector, antisymmetrisation gives schematically

$$D_\mu E_\nu - D_\nu E_\mu = [D_\mu, D_\nu]\Theta = \mathcal{R}_{\mu\nu}(\Omega)\Theta;$$

5. derivation of Einstein dynamics (or a definite UBT modification) from the same action.

These are dynamical questions, not a shortage of local metric components.

Proof status

Proved: central Lorentzian anticommutator identity; local tetrad representation of every Lorentz metric; rank ten of the tetrad-to-metric map; flat inertial limit $\Omega = \Gamma = 0$ when curvature is zero.
Open: derivation of the admissible Lorentz slice and the connection from the single fundamental field/action; global integrability; on-shell representation of all required GR solutions; Einstein dynamics from the canonical UBT action.
Noncanonical alternative: compact-ψ fiber averaging remains a mathematically valid candidate completion but is not used in this theorem and is not part of the minimal covariant-tetrad core.